Diamond Price Prediction

Column Name	Description
carat	The weight of the diamond, measured in carats. A higher carat value means a heavier diamond
cut	Refers to how well the diamond has been cut from its raw form. Categories can range from 'Ideal' (best) to 'Fair' (worst)
color	A measure of how colorless the diamond is. This is graded from 'D' (colorless) to 'J' (light yellow or brown)
clarity	Refers to the absence of inclusions and blemishes. 'IF' means Internally Flawless, indicating no inclusions and only blemishes are visible to a skilled grader. 'I1' means inclusions are obvious under 10x magnification and can affect transparency and brilliance
x	Length in mm
y	Width in mm
z	Depth in mm
depth	Total depth percentage = z / mean(x, y) = 2 * z / (x + y) (43--79)
table	Width of top of diamond relative to widest point (43--95)
price	Price in US dollars (326--18,823)

◇ Imports

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import  missingno as msno
import seaborn as sns
```

◇ Load Dataset

```
df = pd.read_csv('diamonds.csv')
df.head()
```

```
   carat      cut color clarity  depth  table  price     x     y     z
0   0.23    Ideal     E     SI2   61.5   55.0    326  3.95  3.98  2.43
1   0.21  Premium     E     SI1   59.8   61.0    326  3.89  3.84  2.31
```

```
2   0.23     Good     E     VS1   56.9   65.0    327  4.05  4.07  2.31
3   0.29  Premium     I     VS2   62.4   58.0    334  4.20  4.23  2.63
4   0.31     Good     J     SI2   63.3   58.0    335  4.34  4.35  2.75
```

Getting information about the dataset

```
df.shape
```

```
(50000, 10)
```

```
df.dtypes
```

```
carat      float64
cut         object
color       object
clarity     object
depth      float64
table      float64
price        int64
x          float64
y          float64
z          float64
dtype: object
```

```
df.describe()
```

```
               carat         depth         table         price
x  \
count  50000.000000  50000.000000  50000.000000  50000.000000
50000.000000
mean       0.799444     61.753006     57.457830   3944.805440
5.734403
std        0.475173      1.431088      2.232092   3997.938105
1.123077
min        0.200000     43.000000     43.000000    326.000000
0.000000
25%        0.400000     61.000000     56.000000    951.000000
4.710000
50%        0.700000     61.800000     57.000000   2410.000000
5.700000
75%        1.040000     62.500000     59.000000   5351.000000
6.540000
max        5.010000     79.000000     95.000000  18823.000000
10.740000

                   y             z
count  50000.000000  50000.000000
mean       5.737956      3.541056
std        1.145579      0.707065
min        0.000000      0.000000
```

```
25%        4.720000      2.910000
50%        5.710000      3.530000
75%        6.540000      4.040000
max       58.900000     31.800000
```

Getting the unique values of categorical features

```
for column in df.columns:
    if df[column].dtype == 'object':
        print(f"Column: {column}")
        print(df[column].unique())
```

```
Column: cut
['Ideal' 'Premium' 'Good' 'Very Good' 'Fair']
Column: color
['E' 'I' 'J' 'H' 'F' 'G' 'D']
Column: clarity
['SI2' 'SI1' 'VS1' 'VS2' 'VVS2' 'VVS1' 'I1' 'IF']
```

◇ Exploratory Data Analysis (EDA)

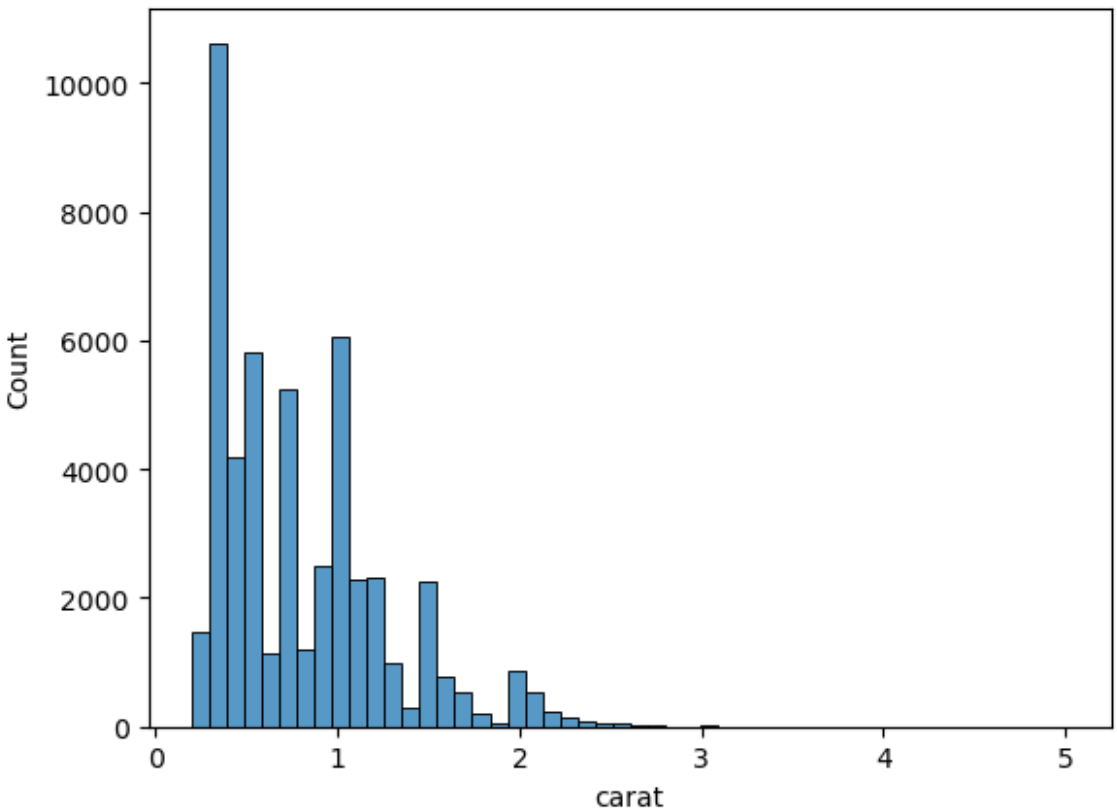
Single Variable Analysis

Carat

Plotting histogram for feature 'carat' to get a good understaning of how many carats the majority of diamonds in the df have

```
sns.histplot(x='carat', data=df, bins=50)
```

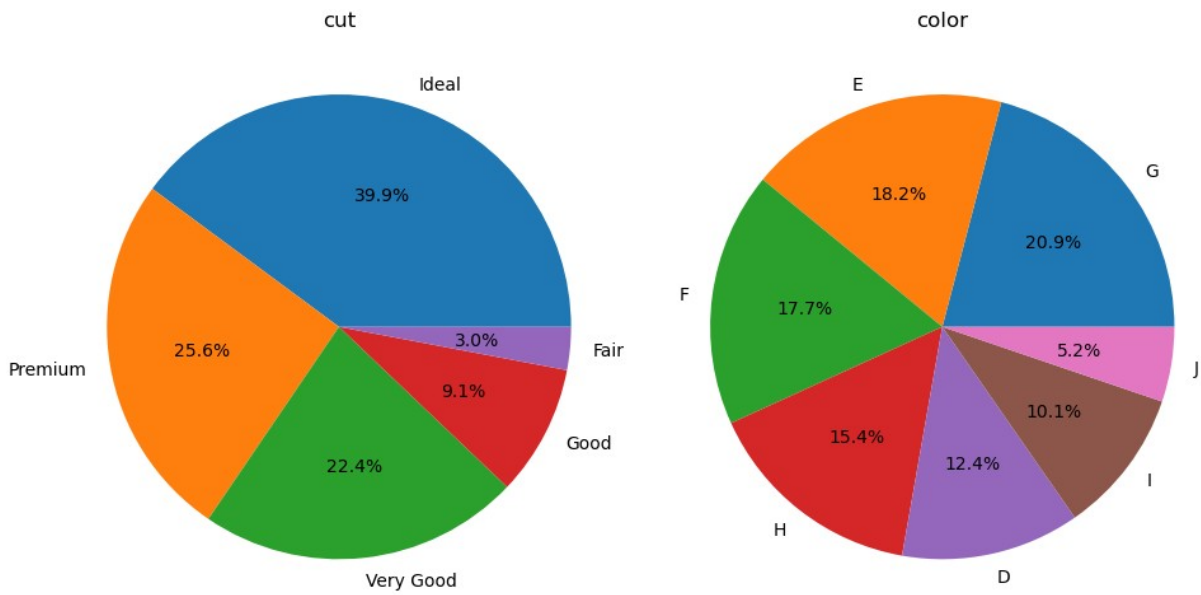
```
<Axes: xlabel='carat', ylabel='Count'>
```



Cuts and Colors

Using pie plot to see how the different types of cuts are distributed among the various diamonds in df and what color the diamonds in the dataset have

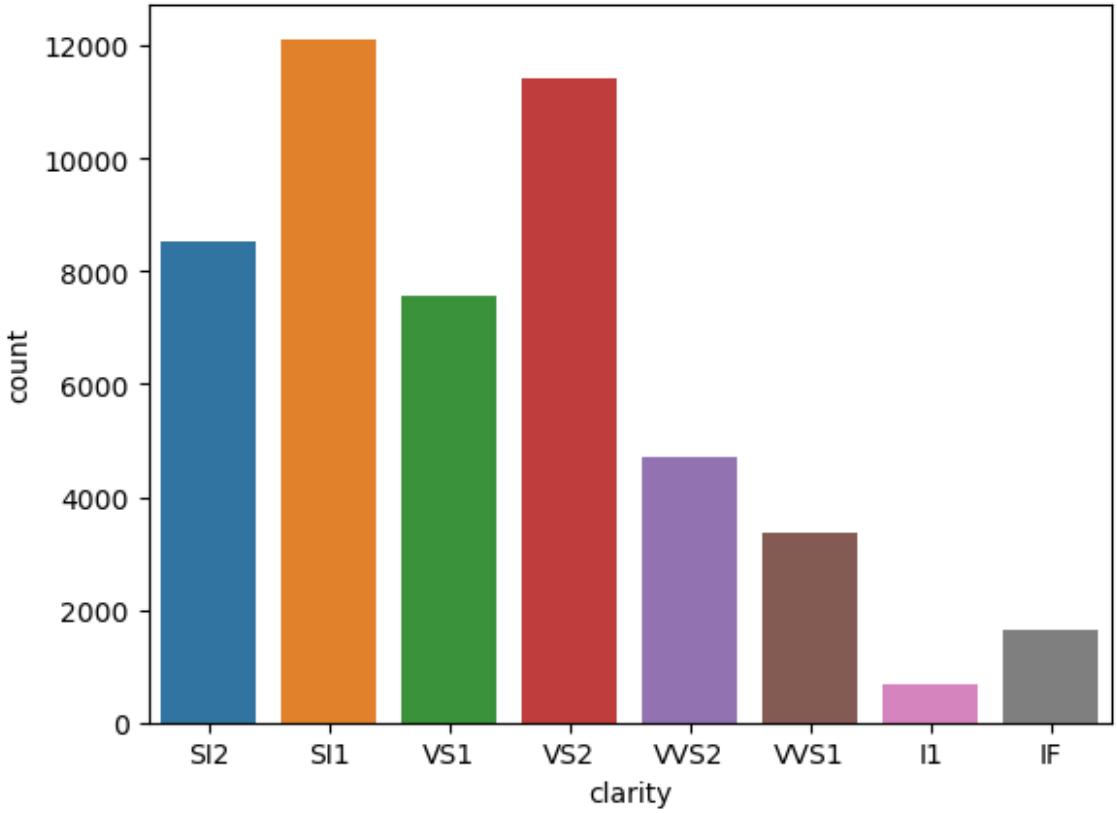
```
cols = ['cut', 'color']

fig, axes = plt.subplots(1, len(cols), figsize=(10, 5))

for i, col in enumerate(cols):
    values = df[col].value_counts()
    axes[i].pie(values, labels=values.index, autopct='%1.1f%%')
    axes[i].set_title(col)

plt.tight_layout()
plt.show()
```



Clarity

```
sns.countplot(x='clarity', data=df)
```

```
<Axes: xlabel='clarity', ylabel='count'>
```



Two Variable Analysis

Cut, Color and Clarity vs Price

Cut:

- The "Ideal" category has the highest frequency, followed by "Very Good" and "Premium." The "Fair" category has the lowest frequency.

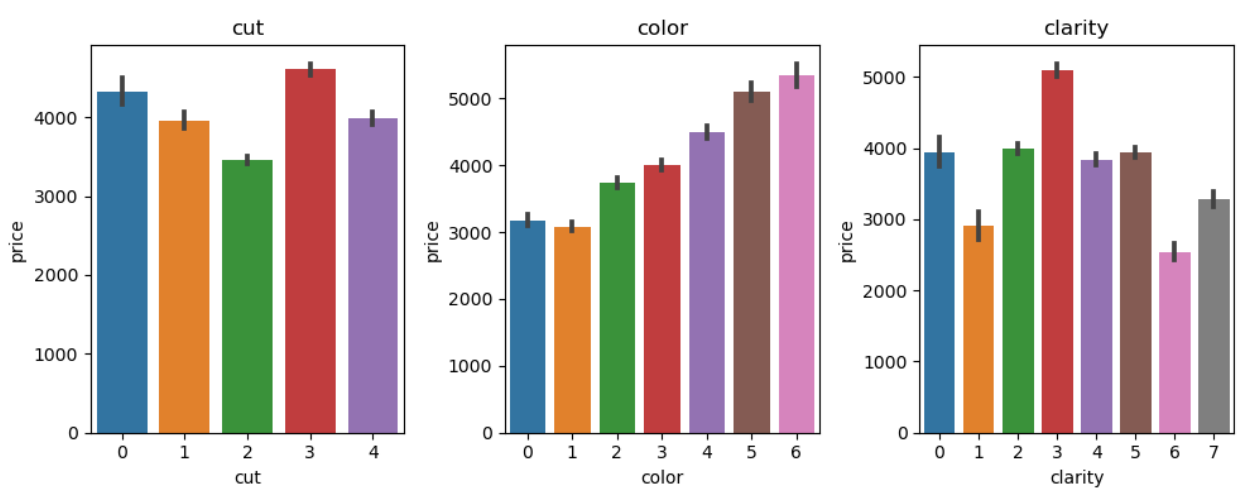
Color:

- The second bar plot represents the color quality of diamonds. It also has five categories: "D," "E," "F," "G," and "H."
- The "G" category seems to be the most common, while "D" and "E" are less frequent.

Clarity:

- The third bar plot shows the clarity of diamonds. It includes categories like "IF" (Internally Flawless), "VVS1" (Very, Very Slightly Included 1), "VS1" (Very Slightly Included 1), and others.
- The "VS1" category appears to be the most prevalent, followed by "SI1" (Slightly Included 1).

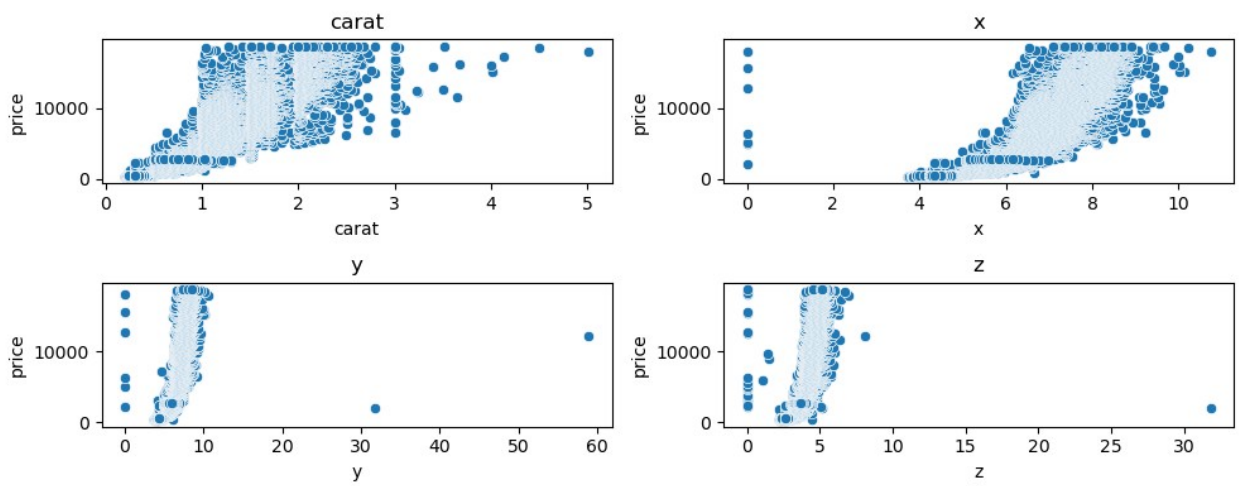
```
cols = ['cut', 'color', 'clarity']

fig, axes = plt.subplots(1, 3, figsize=(10, 4))

for i, col in enumerate(cols):
    ax = axes[i]
    sns.barplot(x=col, y='price', data=df, ax=ax)
    ax.set_title(col)

plt.tight_layout()
plt.show()
```



Carat, X, Y and Z vs Price

As carat weight increases, the price tends to increase as well. This positive correlation suggests that larger diamonds are generally more expensive.

```
cols = ['carat', 'x', 'y', 'z']

fig, axes = plt.subplots(2, 2, figsize=(10, 4))

for i, col in enumerate(cols):
    ax = axes[i//2, i%2]
    sns.scatterplot(x=col, y='price', data=df, ax=ax)
    ax.set_title(col)

plt.tight_layout()
plt.show()
```



◇ Data Preprocessing

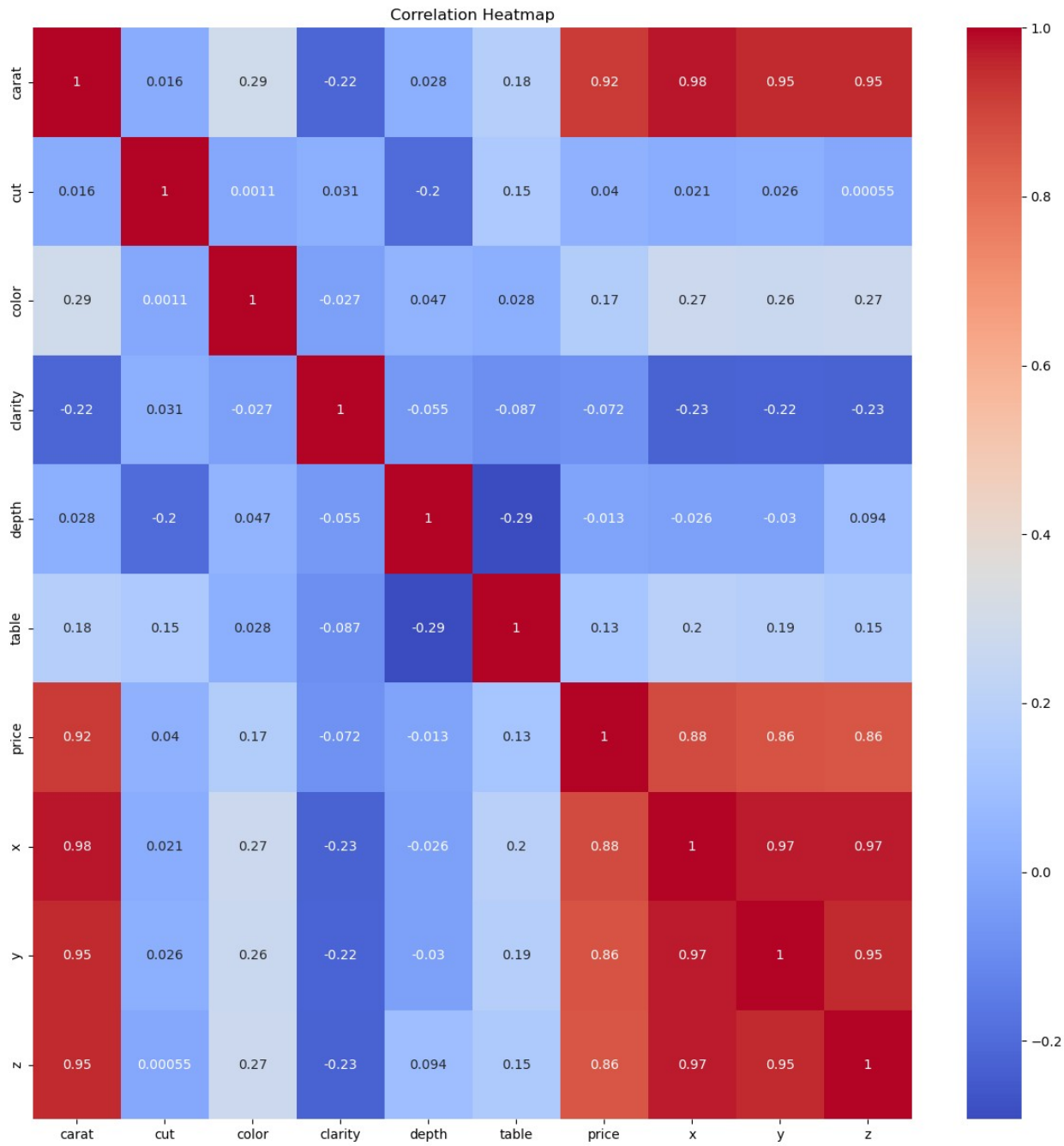
```
from sklearn.preprocessing import LabelEncoder

cols = df.columns[df.dtypes == 'object']

le = LabelEncoder()

for col in cols:
    le.fit(df[col])
    df[col] = le.transform(df[col])

df.dtypes
```

```
carat      float64
cut          int32
color        int32
clarity      int32
```

```
depth      float64
table      float64
price        int64
x          float64
y          float64
z          float64
dtype: object
```

Confusion Matrix

```
plt.figure(figsize=(15, 15))
sns.heatmap(df.corr(), annot=True, cmap='coolwarm')
plt.title('Correlation Heatmap')
plt.show()
```



Train Test Split

```
from sklearn.model_selection import train_test_split

y = df['price']
X = df.drop(columns='price')

x_train, x_test, y_train, y_test = train_test_split(X, y,
test_size=0.2, random_state=42)
```

◇ Model Training

```
from sklearn.tree import *
from sklearn.ensemble import *
from sklearn.ensemble import RandomForestRegressor
from sklearn.neighbors import KNeighborsClassifier
```

Decision Tree

```
dtree = DecisionTreeRegressor(max_depth=4, random_state=42,
ccp_alpha=0.001)

dtree.fit(x_train, y_train)

DecisionTreeRegressor(ccp_alpha=0.001, max_depth=4, random_state=42)

d_pred = dtree.predict(x_test)
print('Training Accuracy: ', dtree.score(x_train, y_train))

Training Accuracy:  0.9067267851814101
```

Random Forest

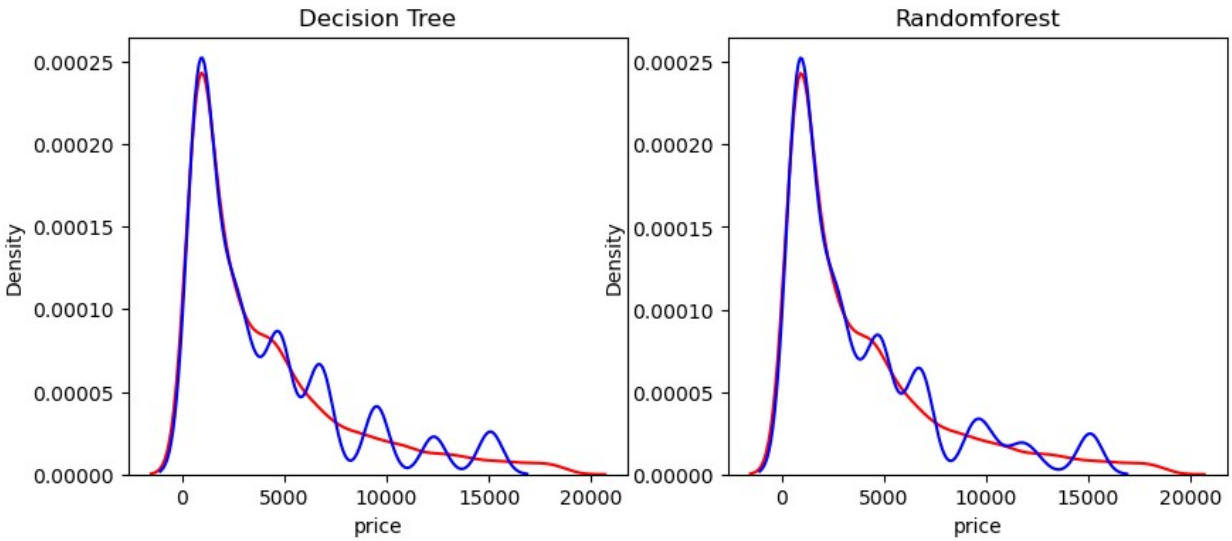
```
rfr = RandomForestRegressor(max_depth=4, random_state=42,
ccp_alpha=0.001)

rfr.fit(x_train, y_train)

RandomForestRegressor(ccp_alpha=0.001, max_depth=4, random_state=42)

rfr_pred = rfr.predict(x_test)
print('Training Accuracy: ', rfr.score(x_train, y_train))

Training Accuracy:  0.9142371609663562
```

◇ Evaluation

```
from sklearn.metrics import accuracy_score, mean_squared_error,
mean_absolute_error, f1_score
```

Distribution Plot

```
models = {'Decision Tree' : d_pred, 'Randomforest': rfr_pred}
fig, ax = plt.subplots(1, 2, figsize=(10, 4))

for i, model_name in enumerate(models):
    sns.kdeplot(y_test, color='r', label="Actual Value",
ax=ax[i]).set_title(model_name)
    sns.kdeplot(models[model_name], color='b', label="Fitted Values" ,
ax=ax[i])
```


Decision Tree
Randomforest
Density
price
0.00000
0.00005
0.00010
0.00015
0.00020
0.00025
0
5000
10000
15000
20000